

LAB ASSIGNMENT - 02

Exploring packet sniffer and packet Analyzer:-

Question 01:-

What are the different protocols you observe at the following layers of the protocol stack?

Solution:- (A) Application layer.

- HTTP
- HTTPS / TLS (for cornell)
- DNS.

B) Transport layer.

- TCP (mostly)
- UDP (for DNS)

C) Network layer

- IP (IPv4/IPv6)

Question-2: What is the total amount of data being received for the following two cases?

A. When you access <http://neverssl.com/>

B. When you access <https://www.cornell.edu>.

Solution: -

A) While accessing `http://neverssl.com/` the total data received are 1486 bytes.

B) While accessing `https://www.corrall.edu` the total data received are 1514 bytes.

Question 03: -

Answer the following when you access the site `http://neverssl.com/`.

A) How many HTTP net request do you observe?
List down the net requests.

Soln: There are 3 HTTP net requests.

- NET | online HTTP/1.1
- NET | online HTTP/1.1
- NET | favicon.ico.

B) What version of HTTP the server is running?

Soln: HTTP/1.1

C) For each of the HTTP net request find out.

(i) the total number of TCP segments being received

Soln: 33 TCP packets.

ii) the total amount of data being received in the corresponding HTTP response message.

Soln $\rightarrow$ HTTP/1.1 200 OK.

Content-length = 680704

1) Identify TCP three way handshake used to establish the connection with the http://newerssl.com. specify syn, syn-ack, Ack packets and write the sequence numbers for each.

Soln :

Packet

Sequence Number.

SYN

$\rightarrow$ 34.104.35.135

SYN-ACK

$\rightarrow$ 10.85.19.186

ACK.

$\rightarrow$ 34.104.35.123.